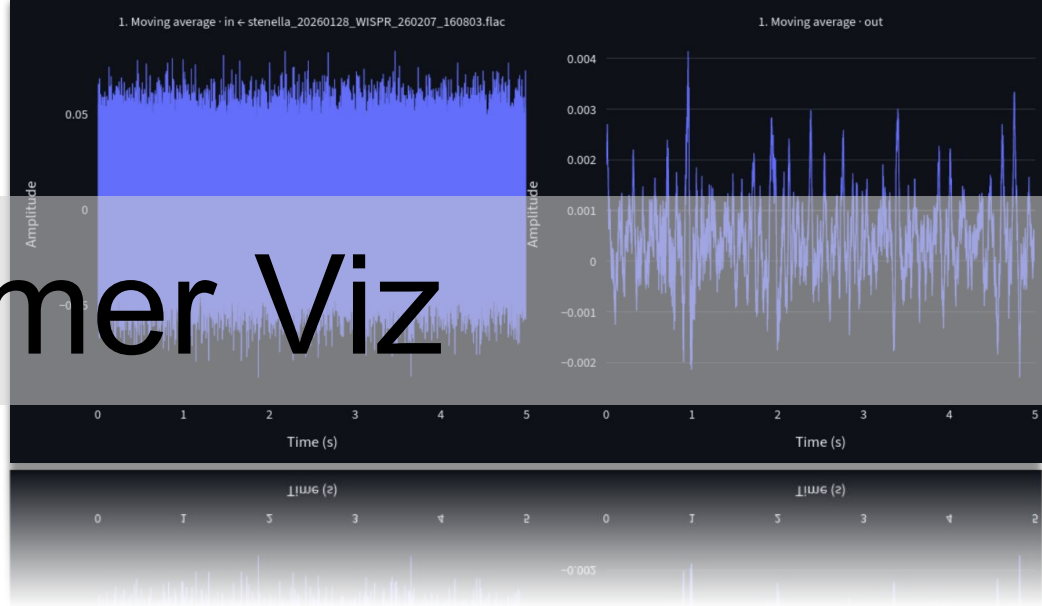


PAM Transformer Viz



A tool for visualizing PAM data processing

Aaron Mau (VOTO)
Brijonnay Madrigal (UCSD)
Ivia Closset (FMI)

PAM Glider Rodeo HackWeek 2026

Background/Justification

- Much of us are new to PAM
- Approaching this from a data analyst perspective
 - How do I clean the data?
 - What is typically done?
- Build familiarity with visualization tools in Python
- Applications to education and data pipelines
- Curiosity was fed through the week
 - Adding additional visualization methods



Methods

- Attempted visualization without `streamlit` and was extremely slow
 - Large datasets, often needing to be cut up
- Brainstorm ideas
 - Base app:
 - *app.py* should handle the user-facing stuff - this is where `streamlit` can come in
 - *data.py* should handle the i/o - loading the files as a dataframe
 - *plots.py* should handle everything with `plotly` for interactivity
 - *transformers.py* should serve as a list of all our known “modifiers” of the data
 - Adding LTSA support
 - Almost like a plugin to in *app.py* that will provide context for the audio files using *ltsa_panel.py* and *ltsa.py*
 - Added playback capabilities
 - *playback.py* allows you to listen to an uploaded sound clip
 - Chaining steps together
 - Another plugin-like feature that could handle multiple transformations in context
- Thinking about our real-world applications

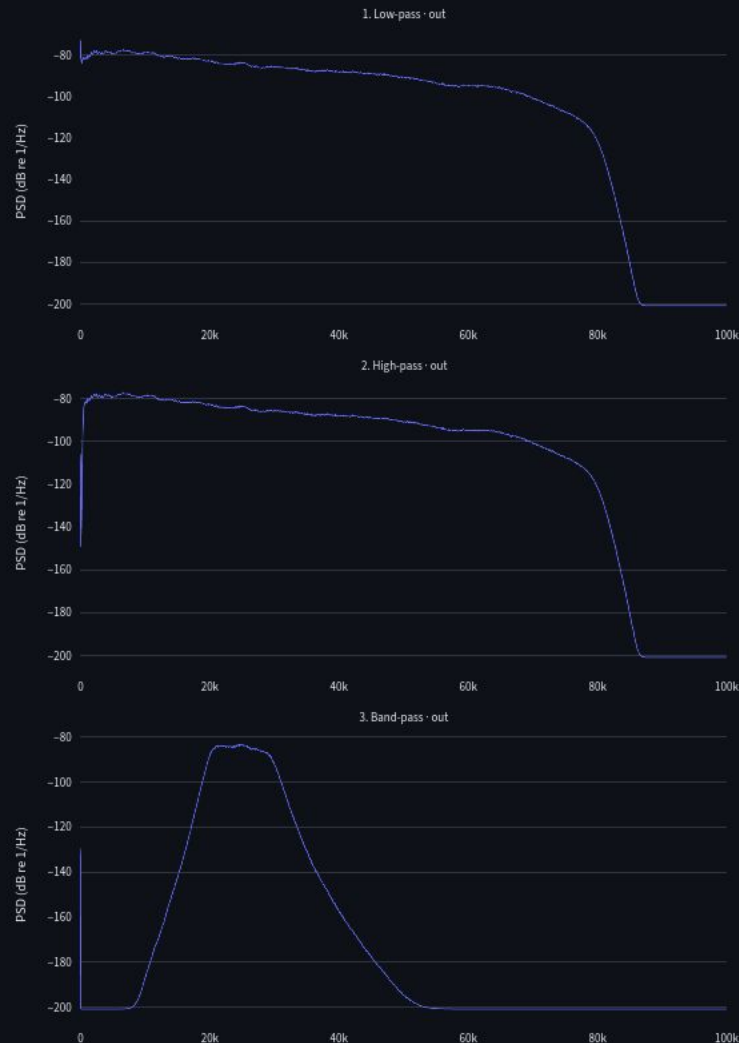
Methods

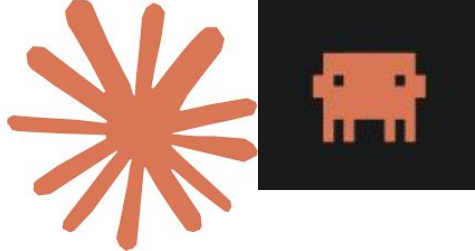
Transformers

- Passthrough
- Low Pass, High Pass, and Band Pass filters
- Notch
- Amplitude normalization
- Envelope
- Downsampling
- FFT and Welch PSD

Plotting types

- Waveform
- Spectrum
- PSD
- Spectrogram





Use of AI

Claude was used for:

- Suggesting filter types and settings
- Creating the code framework and making tweaks

We acknowledge:

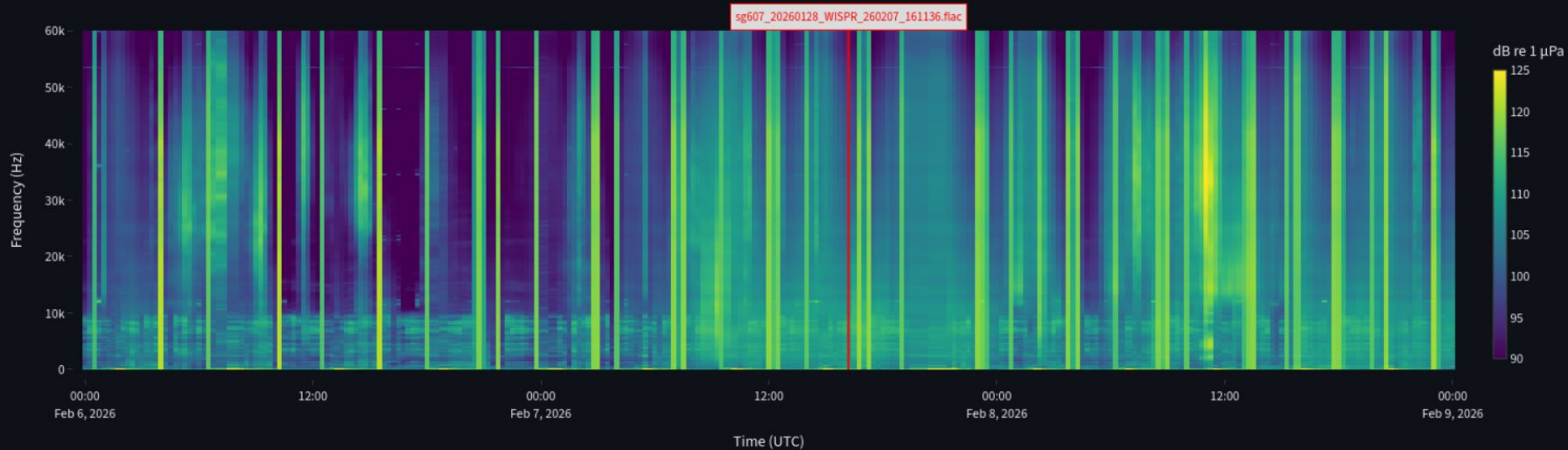
- The code is still being reviewed
- Scientific accuracy needs to be vetted against real data
- Resources were used (energy, water, computing time) to build a tool quickly
- It needs to be documented and maintained *by a human*

“The only difference between screwing around and science is writing it down” - Adam Savage

Results

Part 1: LTSA overlap

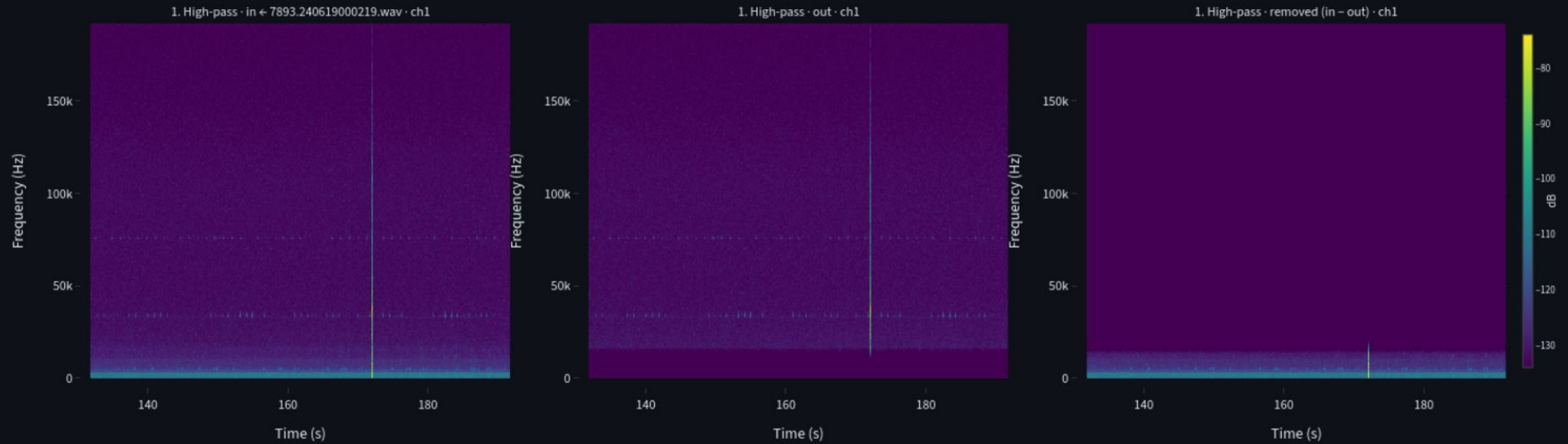
LTSA context



Red marks `sg607_20260128_WISPR_260207_161136.flac` at 2026-02-07 16:11:36 UTC – 16:11:41. LTSA: `sg607_20260128_rodeo_example_5s_100Hz`, 15min averaging, 273 time bins, colour 90–125 dB (fixed).

Results

Part 2: Long, raw data



Lessons Learned

- Aaron
 - How the heck do I even *look* at PAM acoustic data
 - What's a good, *fast* method for loading lots of data
 - What are some other toolboxes out there
- Brijonnay
 - new to gliders
 - new tools (streamlit)
 - navigating new platforms (JupyterLab)
- Ivia
 - New to PAM
 - Working with very large datasets

Next Steps

What are next steps?

- Glider data
 - Does the upcast appear different than the downcast?
 - What about the individual glider states?
- Look at more data
- Documentation
 - Assess scientific validity of the methods
 - A more detailed notebook for the classroom
 - Literature review
- Glider Rodeo - a potentially helpful tool when all the data are released

Please send us your scathing criticisms! [#transformer_vis](#)

Acknowledgements



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Ivia Closset (FMI)



BY ANTHROPIC

**Thank you Glider Rodeo Organizers for
leading this great week!**